Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Under the Auspices of the IAP Action Plan 2022

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Atopic Dermatitis

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Atopic Dermatitis (AD)

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Definition

Atopic dermatitis, is an inflammatory, chronically relapsing, non-contagious, and extremely pruritic skin disease. (WAO)AD affects roughly 20% of the paediatric population.

Clinical features of AD

Major features	Minor features
Must have 3 or more features	***Must have 3 or more following minor features***
Pruritus	Early age of onset, xerosis, palmar hyperlinearity, ichthyosis, keratosis pilaris
Characteristic morphology and distribution	Immediate skin test reactivity, elevated serum IgE, cutaneous infection, including *Staphylococcus aureus* and *Herpes simplex virus*
Chronic or relapsing course	Nipple eczema, cheilitis, pityriasis alba, white dermatographism, delayed blanching, perifollicular accentuation, anterior subcapsular cataracts
Personal or family history of atopy, including asthma, allergic rhinitis, atopic dermatitis	Itch when sweating, non-specific hand or foot dermatitis, Recurrent conjunctivitis, Dennie-Morgan folds, keratoconus, facial erythema or pallor

Scoring for Atopic Dermatitis (Scorad)

A **Area** follow rule of 9.

B **Intensity:** Redness, swelling, oozing/crusting, scratch marks, lichenification, dryness.

C **Subjective symptoms:** Itch and sleeplessness- each scored by the patient or relative using a visual analogue scale where 0 is no itch or sleeplessness and 10 is worst imaginable itch or sleeplessness. These scores are added to give "C" (maximum 20).

Total score (SCORAD) for any individual is A/5 + 7B/2 + C.
If SCORAD is > 50, it indicates severe disease and if SCORAD is < 25, it indicates mild disease.
(European Task Force on Atopic Dermatitis in 1993)

Treatment

1. Non-pharmacological Measures

A **Cleansing and Bathing**

- ☑ Regular once-daily bathing with warm (27–30°C) water of short duration (5–10 minutes)
- ☑ Limited use of non-soap cleansers that are neural to low pH, hypoallergenic and fragrancefree (Syndets)
- ☑ **Bleach baths:** In 0.005% Sodium Hypochlorite can be used for prevention of bacterial colonisation in moderate to severe cases of AD.

B **Moisturizers/Emollients**

- ☑ Prompt, frequent and liberal use of preservative-free and fragrance-free moisturizers.
- ☑ **Soak and seal:** Soak the skin in warm water for 15 minutes, light pat dry and seal in moisturizer for best results. Use atleast 2–3 times a day. Can use "wet wrap therapy" in case of severe flare-ups.

C **Clothing**

Smooth clothing, which is light weight, loose and comfortable, like cotton, is recommended. Wool and synthetic clothing should be avoided.

D **Allergen/Trigger Avoidance**

(As they increase the skin barrier dysfunction)

- ☑ Aeroallergens like pollens and house dust mites should be avoided in allergen sensitive individuals (proven on skin testing). Rooms should be well ventilated with good sunlight, have comfortable temperature, should be clutter free and with minimal upholstery. Should avoid dry dusting and encourage wet mopping.
- ☑ Tobacco smoke avoidance, traffic exhaust and volatile organic compounds exposure reduction (avoid burning wood/ essence sticks/ mosquito repellents) is recommended.

E **Dietary Intervention**

Dietary restriction is recommended in only those individuals with a known food allergy for specific food items.

2. Pharmacological Measures: (Topical)

A Topical Corticosteroids (TCS)

- ☑ **For acute flare-ups (reactive therapy):** Twice daily application till active lesions subsides. To review after 2 weeks. If lesions have come under control in 2 weeks, step down the strength of steroids.[1]
- ☑ **For maintenance (proactive therapy):** Twice weekly application to prevent relapses. (can be used upto16 weeks) with liberal use of emollients. To be applied in well hydrated skin. Lowest potency steroid should be used, suitable for that age.

B Topical Calcineurin Inhibitors (TCI)

Pimecrolimus 1% cream[2] and Tacrolimus[3] 0.03% and 0.1% ointments are effective in both flareups and maintenance.

3. Other Measures

- ☑ **Phototherapy:** In resistant cases.
- ☑ **Antibiotics:** Whenever there is skin infection.
- ☑ **Oral Glucocorticoids:** Short course of low dose steroids, 0.5 mg/kg/day upto 1 week can be used for acute flare-ups.

Newer Modalities

- ☑ **Immunosuppressive drugs:** (For refractory cases) Cyclosporine A, Azathioprine, Methotrexate, Mycophenolate Mofetil can be used.
- ☑ **Biologicals** like dupilumab.
- ☑ **Phosphodiesterase-4 inhibitors** like crisaborole[4] and apremilast.
- ☑ **Janus Kinase inhibitor (JAK1)** like Abrocitinib[5] can be used in resistant cases.
- ☑ **Allergen specific immunotherapy:** In select patients only, of positive sensitization (mostly with house dust mites).

Stepwise management of atopic dermatitis

Step 1: Dry skin	Step 2: Mild AD (SCORAD <25)	Step 3: Moderate AD (SCORAD 25-50)	Step 4: Severead (SCORAD >50)	Step 5: Uncon-trolled AD
☑ Avoidance of triggers ☑ Emollients, Moisturizers	☑ Weak topical steroids / topical CNIs ☑ Emollients, antiseptics	☑ Proactive therapy: Topical Tacrolimus/ Class II or III topical Glucocorticosteroids ☑ Wet wraps, UV therapy	Immunosuppres-sants: Cyclosporin A, Methotrexate, Azathioprine, Myco-phenolate mofetil	Biologicals: Dupilumab in >12 years age

(AD: atopic dermatitis; SCORAD: scoring for atopic dermatitis)

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